

Industry's Largest On-Chip RAM in ARM® Cortex™ - A9 Microprocessor

The Right Choice for Embedded Imaging Applications



Renesas RZ/A1 series microprocessors (MPUs)

feature high-performance ARM® Cortex™-A9 core operating at 400MHz and a large **10MB on-chip RAM** that can buffer the data needed to drive displays up to WXGA (1280x768) without the need for external DRAM memory.

The RZ/A1 series offers enormous advantages in terms of BOM cost, performance, power consumption, and system design time, making it the right choice for Human Machine Interface and other system-on-chip applications.

Applications

Human Machine Interface (HMI), access control, in-house display, smart DCU, machine vision, medical devices, automotive infotainment, barcode imagers, smart printers, IP camera, white goods.

Features

Up to 10MB on-chip RAM

For code execution/data buffering

128-bit memory bus

With parallel (x4) access to deliver superior memory throughput

Execute-In-Place (XIP) from inexpensive QSPI memory

With three layers of cache

LCD controller to drive up to two independent WXGA displays

For vivid display VDC5 and OpenVG1.1 hardware acceleration

Built-in a range of peripherals:

Jpeg Codec, CMOS camera interface, Encryption Engine, Ethernet AVB, etc.

Package option

176, 208, 256 QFP; 324 BGA

Benefits

Lower BOM Cost

- No external SDRAM or LCD controller
- Simpler voltage regulators
- Reduced PCB layers

Increased Performance

- 1000 DMIPS at 400MHz
- 6x throughput of 333MHz 16-bit DDR2

Decreased Power Consumption

- Fewer board components
- No copying of code from flash to RAM

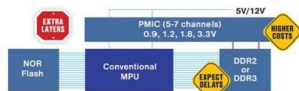
Accelerate Time to Market

- Simpler PCB design
- Decreased EMI via reduced switching
- No DDR procurement issues

Renesas RZ/A1 solution streamlines board design and reduces BOM cost

Conventional Solution

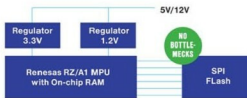
- Complicated power-management and PCB layout
- Memory bandwidth split between code and graphics



BOM Component	Conventional Solution
Flash	\$\$\$ (NOR Flash)
RAM	\$\$\$ (DDR2)
Regulators	\$\$\$ (5-7 PMIC channels)
PCB layers	\$\$\$ (DDR2 supplies, routing)
Total BOM cost	\$\$\$

Renesas RZ/A1 HMI Solution

- Easy system design and testing



BOM Component	RZ/A1 Solution
Flash	\$ (SPI Flash)
RAM	n/a (internal)
Regulators	\$ (3.3V, 1.2V regulator)
PCB layers	\$ (as few as two)
Total BOM cost	\$

Renesas Electronics India Pvt. Ltd.



<http://in.renesas.com/rza1>

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